

The Bruhat–Tits Tree Code

— roadmap handout

Rowan Brad Quni-Gudzinis — QNFO · CWI Summer School poster session, Wed 26 Aug 2026 · poster-v5-treecode candidate · full plan: docs/tree-code-roadmap.md in github.com/QNFO/cwi-qec-poster-2026

The claim

Holographic perfect-tensor codes on a truncated Bruhat–Tits tree. Computed thresholds (analytical recursion + exact stabilizer simulation + Monte Carlo, re-verified by a deposited 35-check script): bit-flip **50.0%** ($4.6\times$ surface), depolarizing **75.0%** ($75\times$), independent $X+Z$ **17.30%**. **No hardware demonstration yet.**

Where the evidence stands

- First falsifiable prediction (log-periodic CMB oscillations) tested → **null** (Planck 2018, bootstrap $p=0.89$; bispectrum upper bounds only).
- Published boundary: under independent errors the qudit family $\approx 2\times 10^{-4}$ ($\sim 55\times$ below surface); the claim targets correlated-failure regimes.
- QEC–Darwinism no-go audited on tree spaces:

boundary moves to $F_L \approx 0.83\text{--}0.85$, residual gap disclosed.

- Self-correction published: naive p-adic valuation reading of stabilizer codes carries no new content.

What we test next

#	Test	Where	Disconfirmation criterion
T1	Matched-noise threshold comparison vs surface	Classical, days–weeks	Advantage falls to parity under matched noise
T2	Decoder complexity + full overhead	Classical	Advantage disappears with decoding cost
T3	Random-walk return probability on p-regular trees	Analog/ion, ~8 wk	No change in decay exponent
T4	Clock-rest ultrametricity split (0% vs 29–35%)	Trapped ion, ~8 wk	Split outside predicted range
T5	Independent re-implementation of T1	Community	Reproducibility test

What we'd like from you

1. An openly specified **correlated-noise benchmark** (model + data) for T1 · 2. decoder-complexity reality check for tree-structured codes · 3. pointers on **perfect tensors for $p > 2$** · 4. an **independent re-run of T1** — the simulator and protocol are handed over.

DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21902891 (Planck null) · 10.5281/zenodo.21901664 (bispectrum bounds) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21964674 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction) · 10.5281/zenodo.21992118 (ZX seam)